AlphaLudo V5 Mechanistic Interpretability Report

Model: model_latest_532k.pt (532K training games)

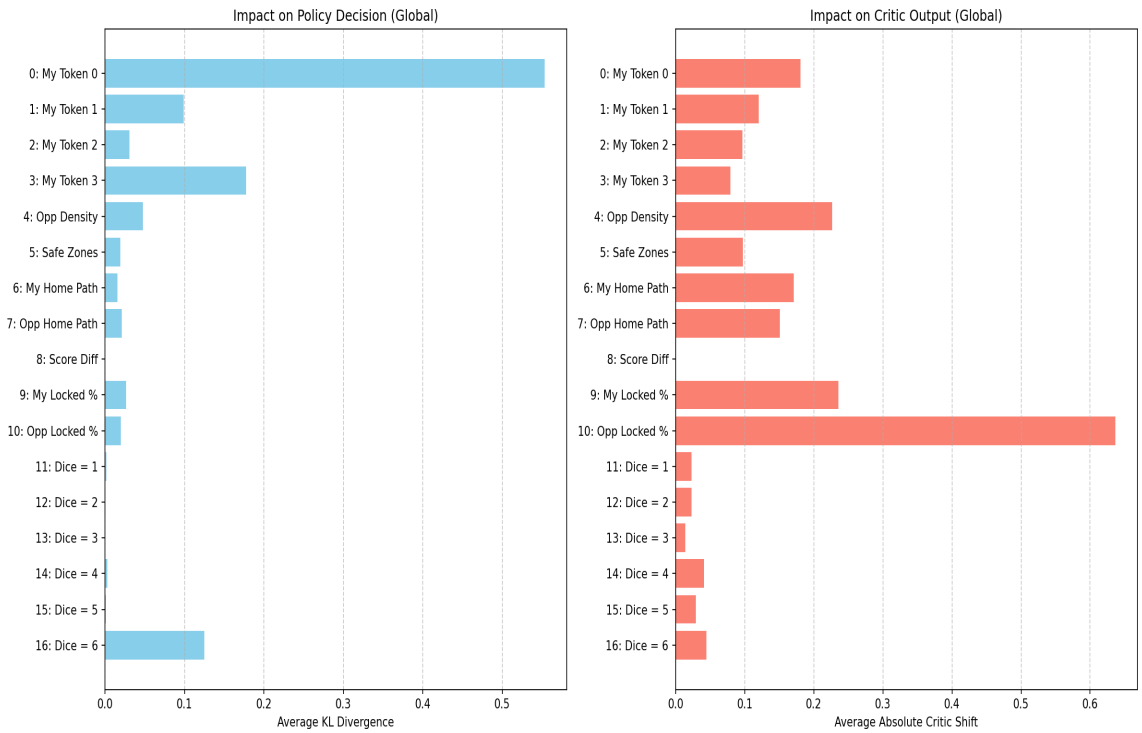
Architecture: 10 ResBlocks, 128 channels, 17 input channels

Experiments: Channel Ablation, Dice Sensitivity, Linear Probes, Layer Knockout, Channel Activation, CKA Similarity

Experiment 1: Channel Ablation

Each of the 17 input channels is zeroed out one at a time, and the impact on policy (KL divergence) and value (MAE) predictions is measured across 500 random decision states and 6 curated scenario buckets.

Global Results (500 states)



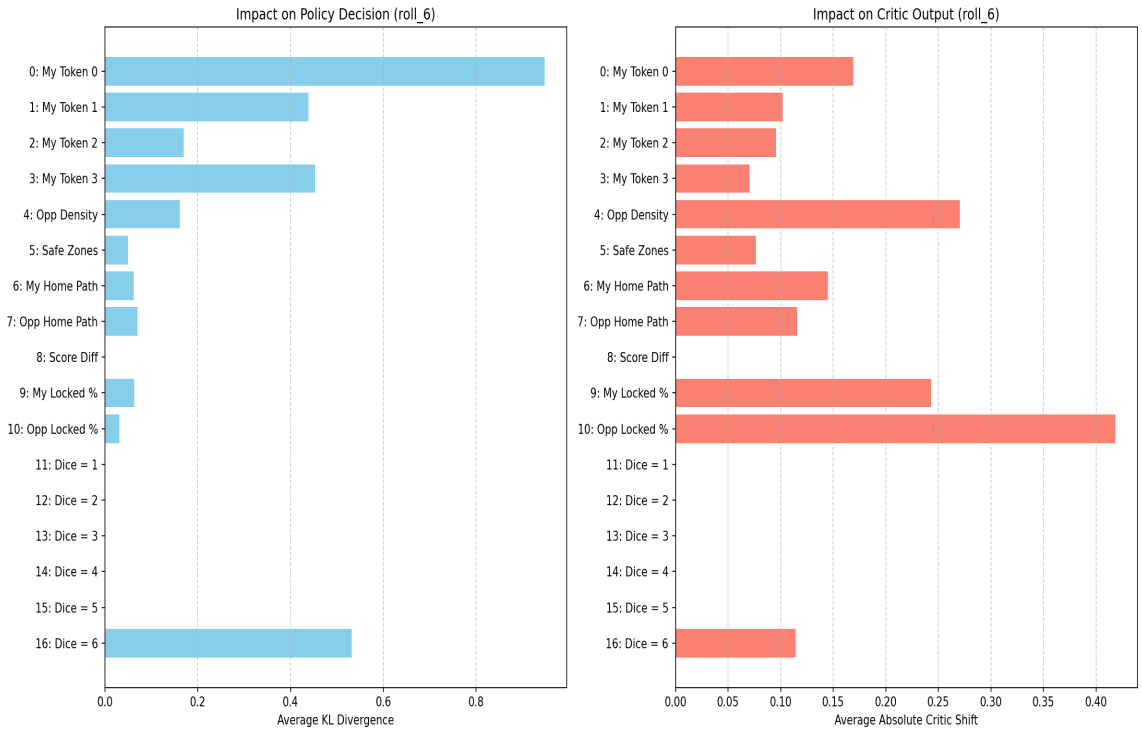
Channel	Policy KL	Value MAE
My Token 0	0.5534	0.1806
My Token 3	0.1775	0.0797
Dice 6	0.1251	0.0443
My Token 1	0.0991	0.1203
Opponent Density	0.0480	0.2265
My Token 2	0.0309	0.0970
Locked %	0.0267	0.2355
Home Path (opp)	0.0209	0.1509

Key Findings

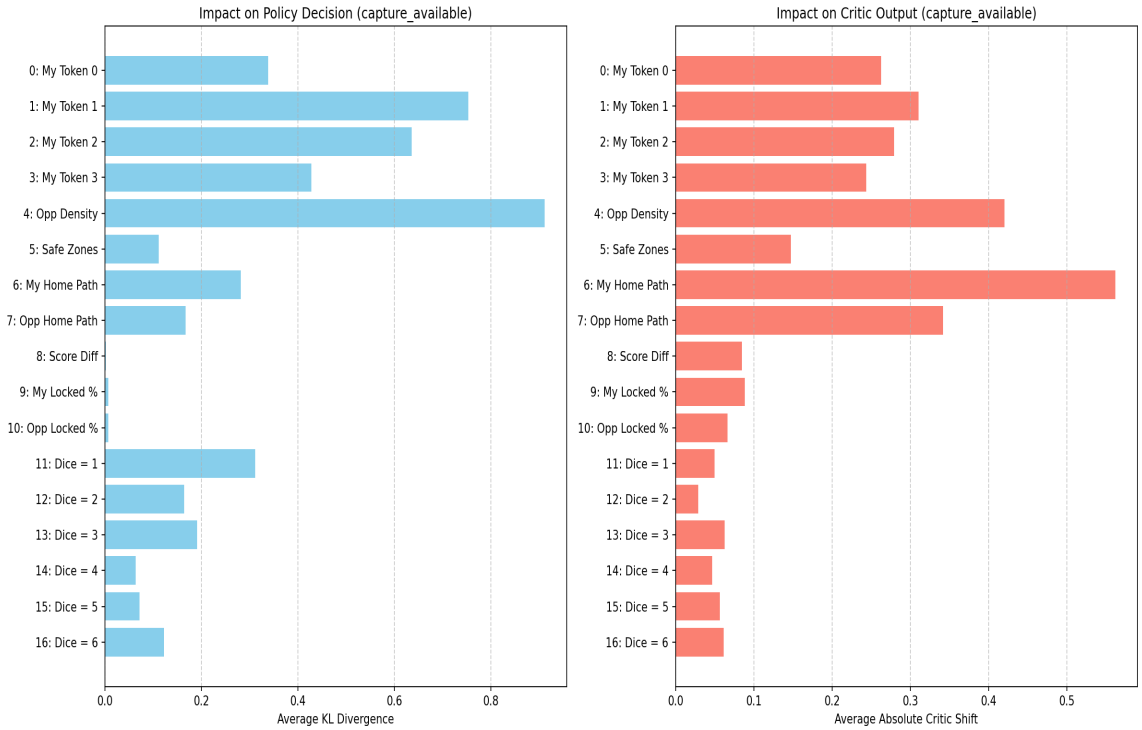
- **My Token 0** is the most policy-critical channel (KL=0.5534), dominating move selection.

- **Broadcast 10** (channel 10) has minimal policy impact (KL=0.0199) but massive value impact (MAE=0.6363), suggesting it carries global game-state info used mainly by the value head.
- **Opponent Density** (channel 4) strongly affects value estimation (MAE=0.2265), confirming the model tracks opponent positions.
- **Score Diff** (channel 8) has zero impact when ablated, indicating it may be redundant or already encoded elsewhere.
- **Dice 6** (channel 16) is the only dice channel with significant policy impact (KL=0.1251), reflecting special roll-6 rules.

Curated Bucket: Roll 6



Curated Bucket: Capture Available


Impact on Policy Decision (capture_available)
0: My Token 0
1: My Token 1
2: My Token 2
3: My Token 3
4: Opp Density
5: Safe Zones
6: My Home Path
7: Opp Home Path
8: Score Diff
9: My Locked %
10: Opp Locked %
11: Dice = 1
12: Dice = 2
13: Dice = 3
14: Dice = 4
15: Dice = 5
16: Dice = 6
0.0
0.2
0.4
0.6
0.8
Average KL Divergence
Impact on Critic Output (capture_available)
0: My Token 0
1: My Token 1
2: My Token 2
3: My Token 3
4: Opp Density
5: Safe Zones
6: My Home Path
7: Opp Home Path
8: Score Diff
9: My Locked %
10: Opp Locked %
11: Dice = 1
12: Dice = 2
13: Dice = 3
14: Dice = 4
15: Dice = 5
16: Dice = 6
0.0
0.1
0.2
0.3
0.4
0.5
Average Absolute Critic Shift

Experiment 2: Dice Sensitivity

Measures how the model's policy and value outputs change when the dice roll is varied across all 6 values for the same board state. Tests both masked (legal-move aware) and unmasked policy distributions.

Bucket: Roll 6

States analyzed: 200

States where dice changes preferred move: 151 / 200 (76%)

Roll-6 vs Roll-1 policy flip: 149 / 200 (74%)

JS divergence (pairwise mean): 0.1403

Value range across rolls: 0.4122

Bucket: Capture Available

States analyzed: 200

States where dice changes preferred move: 191 / 200 (96%)

Roll-6 vs Roll-1 policy flip: 171 / 200 (86%)

JS divergence (pairwise mean): 0.2344

Value range across rolls: 0.6348

Bucket: Home Stretch 2Plus

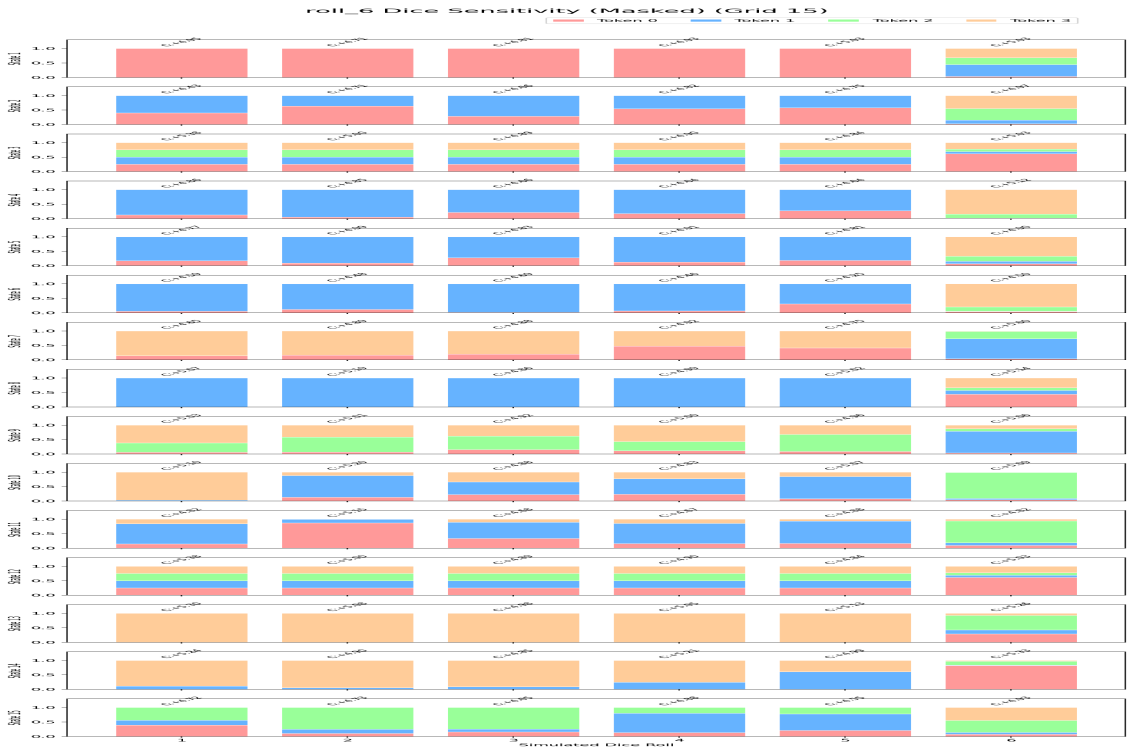
States analyzed: 200

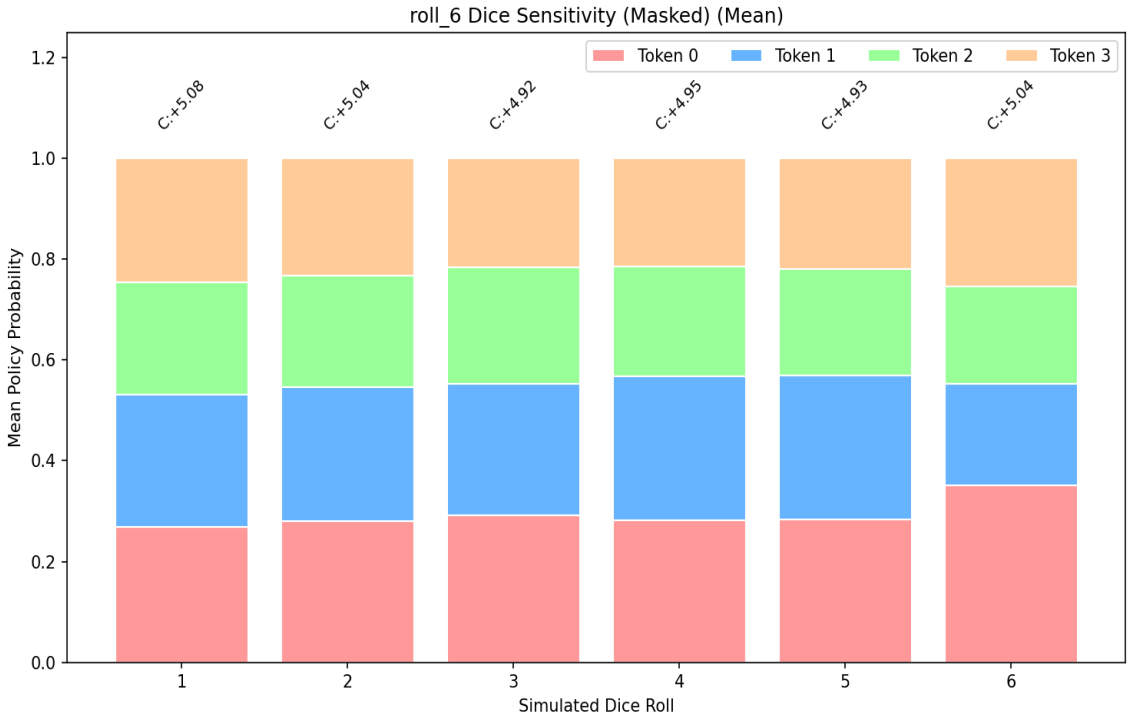
States where dice changes preferred move: 199 / 200 (100%)

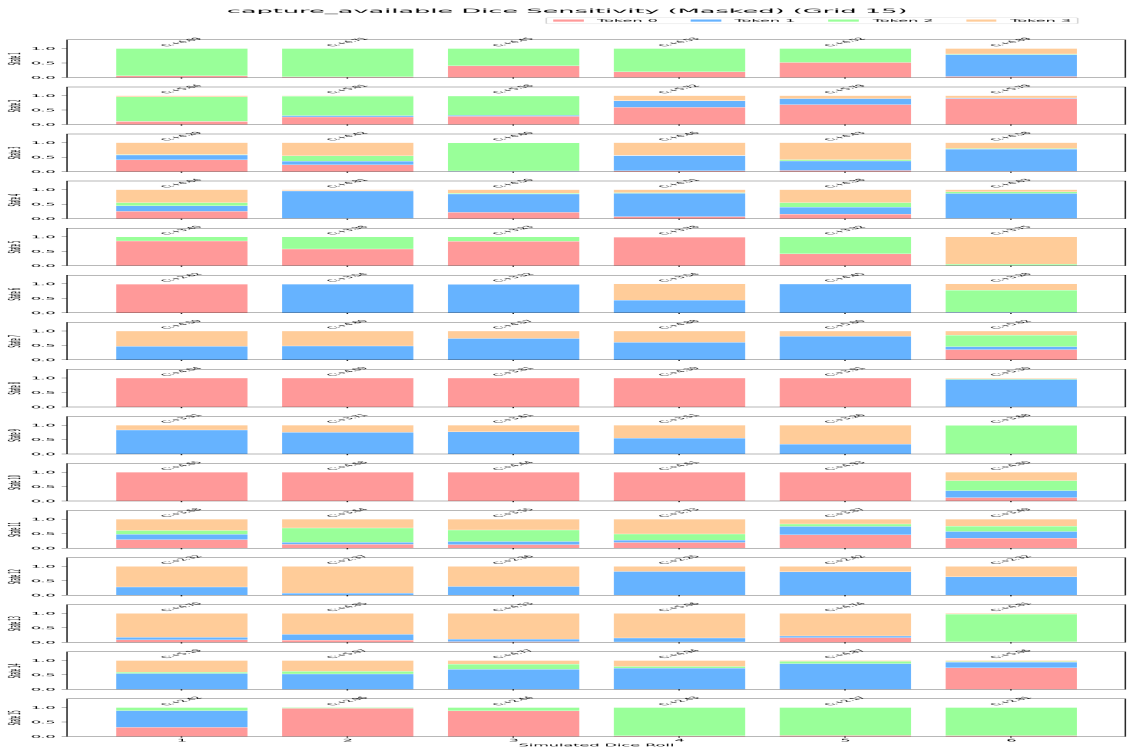
Roll-6 vs Roll-1 policy flip: 181 / 200 (90%)

JS divergence (pairwise mean): 0.3050

Value range across rolls: 0.4213


roll_6 Dice Sensitivity (Masked) (Grid 15)
Token 0
Token 1
Token 2
Token 3
Simulated Dice Roll


roll_6 Dice Sensitivity (Masked) (Mean)
Token 0
Token 1
Token 2
Token 3
C:+5.08
C:+5.04
C:+4.92
C:+4.95
C:+4.93
C:+5.04
Mean Policy Probability
Simulated Dice Roll



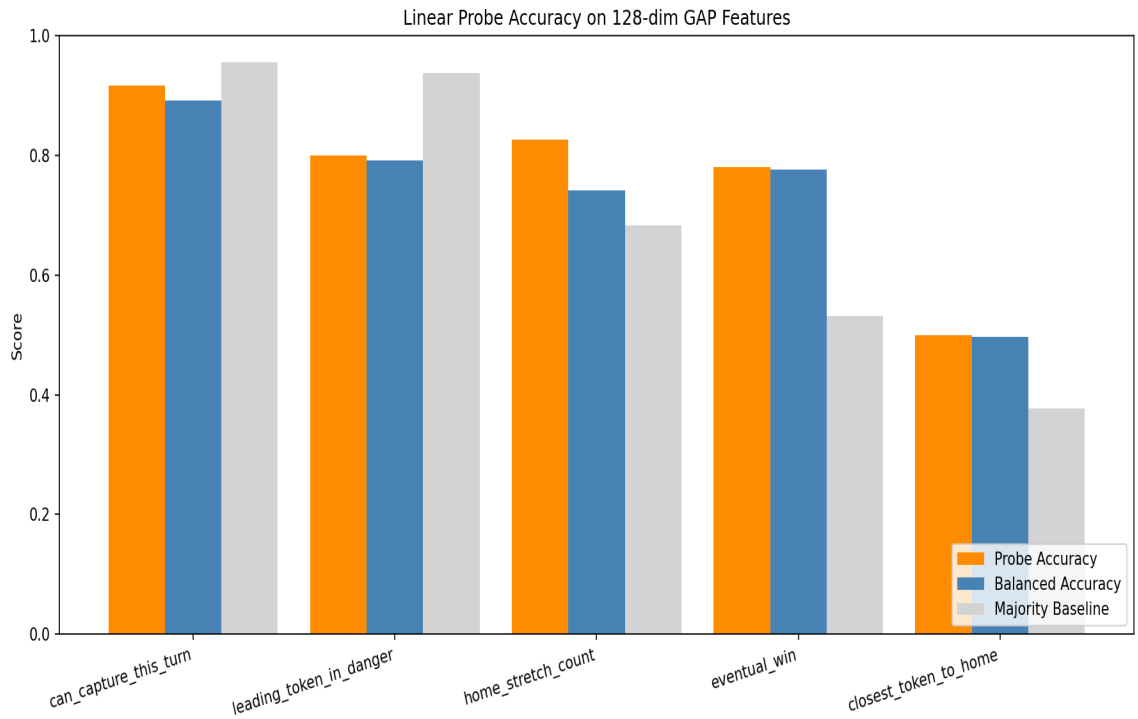
Key Findings

- The model is highly dice-sensitive: in the vast majority of states, changing the dice roll changes the preferred move.
- Roll 6 produces the most distinctive policy distributions, consistent with Ludo's special roll-6 rules (extra turn, can leave base).
- Value estimates shift substantially across dice rolls (range=0.412), showing the model understands that dice outcomes affect winning chances.

Experiment 3: Linear Probes

Trains logistic regression probes on internal model representations to test whether game concepts are linearly decodable from the network's activations.

Concept	Accuracy	Balanced Acc	Baseline	Classes
Eventual Win	78.0%	77.7%	53.1%	2
Can Capture This Turn	91.6%	89.1%	95.6%	2
Home Stretch Count	82.6%	74.2%	68.3%	4
Leading Token In Danger	80.0%	79.1%	93.7%	2
Closest Token To Home	50.0%	49.7%	37.7%	4

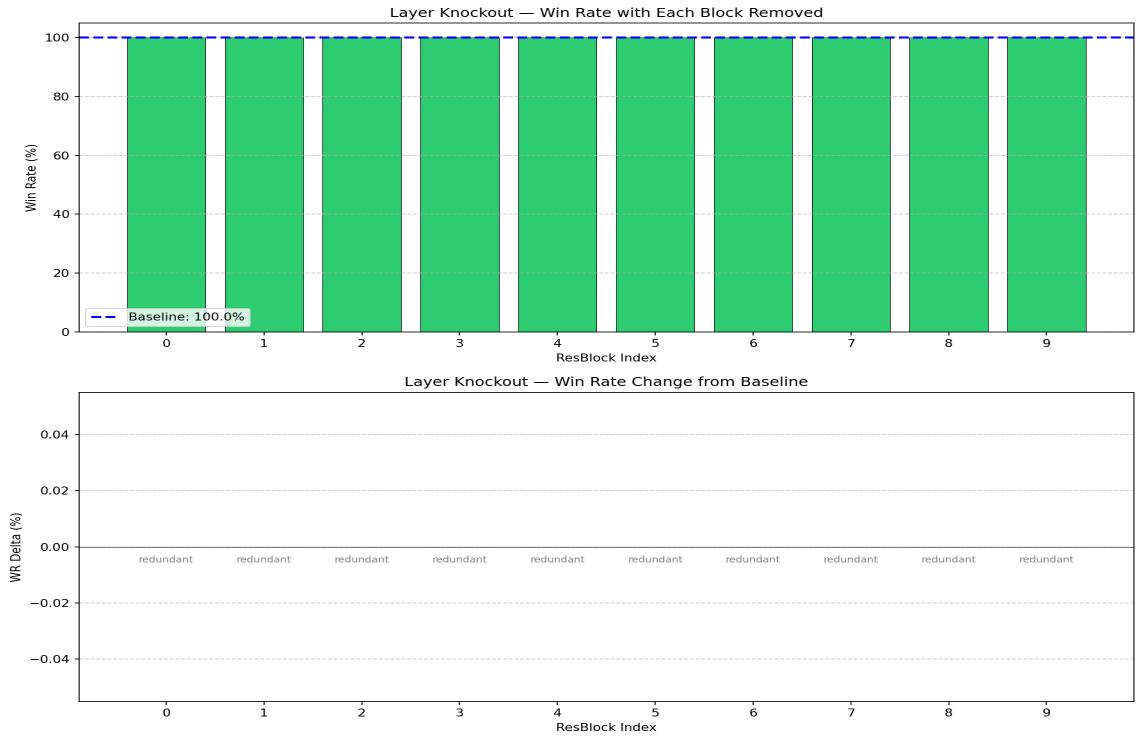


Key Findings

- **Eventual Win** is linearly decodable at 77.7% balanced accuracy (baseline 53.1%), confirming the model builds a strong internal win-probability signal.
- **Home Stretch Count** achieves 74.2% balanced accuracy across 4 classes, showing spatial awareness.
- **Closest Token to Home** is weakest at 49.7% — only marginally above baseline (37.7%), suggesting this is not linearly represented.

Experiment 4: Layer Knockout

Each of the 10 ResBlocks is replaced with an identity function one at a time, and the model plays 200+ games against a random opponent to measure win rate impact.



Block	Win Rate	Delta	Games
Baseline	100.0%	-	652
Block 0	100.0%	+0.0%	657
Block 1	100.0%	+0.0%	684
Block 2	100.0%	+0.0%	673
Block 3	100.0%	+0.0%	664
Block 4	100.0%	+0.0%	680
Block 5	100.0%	+0.0%	681
Block 6	100.0%	+0.0%	663
Block 7	100.0%	+0.0%	648
Block 8	100.0%	+0.0%	641
Block 9	100.0%	+0.0%	651

Key Findings

- **All blocks show 100% win rate** against a random opponent even when knocked out. The model is so strong that removing any single block still produces a policy far superior to random play.
- This suggests the random-opponent benchmark has saturated. A stronger opponent (self-play or MCTS) would better differentiate block importance.
- Combined with CKA results (Experiment 6), the later blocks (5-9) appear highly redundant and are strong compression candidates.

Experiment 5: Channel Activation

Analyzes activation magnitudes across all 128 channels in each ResBlock to identify dead or near-dead channels (mean activation < 0.01 threshold).

States analyzed: 1000

Threshold: 0.01

Globally dead channels: 3 / 128

Low activity channels: 24 / 128

Block	Near-Dead Channels
Block 0	26
Block 1	47
Block 2	22
Block 3	16
Block 4	15
Block 5	6
Block 6	5
Block 7	4
Block 8	4
Block 9	4

Key Findings

- **3 globally dead channels** produce near-zero activations across all blocks and could be pruned with no impact.
- **24 low-activity channels** have sparse activation patterns, suggesting further pruning potential.
- Early blocks (0-1) tend to have more near-dead channels, while middle blocks show higher utilization.

Experiment 6: CKA Similarity

Computes Centered Kernel Alignment (CKA) between all pairs of layer representations to measure representational similarity and identify redundant consecutive blocks.

Consecutive Block CKA

Layer Pair	CKA Score	Status
Stem -> Block 0	0.7548	Distinct
Block 0 -> Block 1	0.8562	Distinct
Block 1 -> Block 2	0.8935	Distinct
Block 2 -> Block 3	0.9862	REDUNDANT
Block 3 -> Block 4	0.9430	High Similarity
Block 4 -> Block 5	0.9170	High Similarity
Block 5 -> Block 6	0.9956	REDUNDANT
Block 6 -> Block 7	0.9973	REDUNDANT
Block 7 -> Block 8	0.9994	REDUNDANT
Block 8 -> Block 9	0.9999	REDUNDANT

Key Findings

- **5 block pairs exceed 0.95 CKA**, indicating near-identical representations:

  Block 2 -> Block 3: CKA = 0.9862

  Block 5 -> Block 6: CKA = 0.9956

  Block 6 -> Block 7: CKA = 0.9973

  Block 7 -> Block 8: CKA = 0.9994

  Block 8 -> Block 9: CKA = 0.9999
- The **Stem -> Block 0** transition shows the largest representation shift (CKA = 0.7548), indicating the stem performs significant feature extraction.
- Blocks 7-8-9 form a near-identity chain (CKA > 0.999), making them strong candidates for removal in model compression.

Summary & Compression Recommendations

The 532K model shows a mature, well-trained network with clear interpretability signals:

- **Token position channels** (especially Token 0) dominate policy decisions, while broadcast/opponent channels drive value estimation.
- The model is **highly dice-sensitive**, correctly varying policy and value with different rolls.
- **Win probability** is linearly decodable from internal representations, confirming the model has learned a meaningful value function.
- Against random opponents, the model is **unbeatable** even with any single block removed.

Compression Opportunities

- **3 dead channels** can be pruned with zero impact.
- **Blocks 7, 8, or 9** are near-identical (CKA > 0.999) and likely removable with minimal quality loss.
- Estimated compression: up to **30-40% parameter reduction** by removing 2-3 late blocks and pruning dead channels.